CatchFire Matching Engine — Status Report

Date: February 23, 2026

Project: cf-influencer-matching-engine

Version: v0.8.0

Node.js: v24 | **Vulnerabilities:** 0 | **Tests:** 119 unit/E2E passing + 13 integration (require server)

System Health

SERVICE	STATUS	DETAILS
Cloud Run	Online	Revision cf-matching-engine-00014-ln5, us-central1
Firestore	Online	creators collection, 37 documents
Gemini AI	Online	gemini-2.5-flash
Embeddings	Online	gemini-embedding-001, 768 dimensions
Golden Records	Active	11 benchmark creators across 8 crafts
GitHub CI	Passing	Unit tests + type check + build + smoke tests
Authentication	Active	Bearer token auth on all mutating endpoints
Claude-Mem	Online	Port 37777, Chroma vector DB on port 8000

Executive Summary

Today's sprint executed a full 14-task development cycle covering every planned "Next Step" from the v0.7.0 release, plus critical infrastructure fixes and cross-project documentation improvements. The sprint began with diagnosing and repairing a non-functional AI memory system (Chroma vector database), then parallelized 10 feature tasks across development agents. Post-sprint, all issues were documented in a new centralized global lessons-learned system deployed across 6 repositories.

Key outcomes:

- **14 of 14 tasks completed** in a single session
- **43 files changed**, 26 new files created
- **41 new tests** added (33 E2E semantic search + 8 fixture integrity)
- **3 new API endpoints** (scraper trigger, status, reports)
- **2 new React pages** (Login, Scraper Dashboard)
- **Authentication system** deployed to production
- **Production redeployed** to Cloud Run (revision 00014)
- **Global LEARN.md system** created for cross-project knowledge sharing

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What Changed Since v0.7.0

Infrastructure Fixes

FIX	IMPACT
Claude-Mem Chroma vector DB repair	AI memory system restored, startup 60s → 1s

FIX	IMPACT
`gh auth` workflow scope added	GitHub Actions workflows can now be pushed
`API_AUTH_TOKENS` + `SCRAPER_API_KEY` set	Production auth enabled
Production redeployed (rev 00014)	All v0.8.0 features live

New Features

FEATURE	DESCRIPTION
ErrorBoundary	Global React error catching with fallback UI
Toast Notifications	Transient error/success feedback (4 types: error, success, warning, info)
ApiError Class	Differentiated network vs. HTTP errors with `isTransient`, `isServerError` getters
Brief Templates	10 pre-built search templates across 5 categories (Film, Brand, Motion, Music, Experiential)
Login Page	Token-based authentication at `/app/login`
Scraper Dashboard	Admin dashboard at `/app/scraper` with trigger controls and status monitoring
Auth Middleware	`requireAuth` (401/403) + `optionalAuth` (non-blocking flag) on all endpoints
Scraper API	`POST /trigger`, `GET /status`, `GET /reports` endpoints for scraper orchestration

Testing Improvements

IMPROVEMENT	DETAILS
E2E semantic search tests	33 new tests across 7 suites (full pipeline, negative, multi-craft, Golden Record boost)
Test data fixtures	12 mock creators, 768-dim embeddings, Golden Record fixtures
Fixture integrity tests	8 tests validating fixture data structure

DevOps & Deployment

ITEM	DETAILS
Cloud Monitoring scripts	`deploy-monitoring.ps1/.sh` — uptime check + 2 alert policies
Cloud Scheduler configs	3 jobs: daily Vimeo, daily Behance, weekly all-platform
Staging environment	`deploy-staging.ps1/.sh`, GitHub Actions workflow, staging docs
CSS token cleanup	17 new design tokens, 21 raw RGBA → CSS variable replacements

Documentation & Knowledge Management

ITEM	DETAILS
LEARN.md	6 new entries (L011–L016) documenting session fixes
LEARN-GLOBAL.md	Centralized cross-project lessons file with 9 entries (G001–G009)
Global Cursor rule	`global-learn.mdc` ensures AI always consults global lessons
6 repos updated	Cross-reference headers added to all project LEARN.md files

Database Statistics

METRIC	VALUE
Total Creators	37
Golden Records	11
Crafts Covered	8 (cinematographer, director, motion_designer, vfx_artist, colorist, animator, sound_designer, 3d_artist)
Platforms	5 (vimeo: 13, other: 18, behance: 3, artstation: 2, youtube: 1)
Embedding Dimensions	768

API Endpoints (24 Total)

Search & Discovery

- `POST /api/v1/search/semantic` — AI-powered semantic search
- `GET /api/v1/lookalikes` — Golden Record lookalike recommendations
- `GET /api/v1/similar/:id` — Find creators similar to a given creator

Creator CRUD

- `GET /api/v1/creators` — List/search/filter creators (optional auth)
- `GET /api/v1/creators/:id` — Get creator by ID (optional auth)
- `POST /api/v1/creators` — Add single creator (Zod validated, **requires auth**)
- `POST /api/v1/creators/batch` — Bulk import (**requires auth**)
- `POST /api/v1/import/apify` — Apify scraper import with LLM categorization (**requires auth**)

AI / LLM

- `POST /api/v1/match` — Match creators to a brief
- `POST /api/v1/categorize` — LLM auto-categorize from bio (**requires auth**)
- `POST /api/v1/style-signature` — Generate style signature (**requires auth**)
- `GET /api/v1/llm/test` — Test LLM connection (optional auth)
- `POST /api/v1/embed` — Generate embeddings for a creator (**requires auth**)
- `POST /api/v1/embed/batch` — Batch embedding generation (**requires auth**)

Scraper Management (NEW)

- `POST /api/v1/scraper/trigger` — Trigger scraper run (**requires auth**)
- `GET /api/v1/scraper/status` — Current scraper status (**requires auth**)
- `GET /api/v1/scraper/reports` — Recent scrape reports (**requires auth**)

System

- `GET /health` — Health check (public)
- `GET /api/v1/stats` — Database statistics (optional auth)
- `POST /api/v1/feedback` — Thumbs up/down feedback (**requires auth**)

Web UI

- `GET /` — Beta Control Center (primary dashboard)
- `GET /dashboard` — Analytics dashboard

- `GET /testing` — Match testing UI
- `GET /app/*` — React SPA (8 pages including new Login and Scraper Dashboard)

Web UI Pages (10 Total)

PAGE	URL	STATUS
Beta Control Center	`/`	Updated (brief templates, error handling)
How It Works	`/app/how-it-works`	Unchanged
Browse Creators	`/app/creators`	Unchanged
Creator Profile	`/app/creators/:id`	Updated (fixed silent errors, toast feedback)
Admin	`/app/admin`	Unchanged
Status	`/app/status`	Unchanged
Login	`/app/login`	**NEW** — Token-based authentication
Scraper Dashboard	`/app/scraper`	**NEW** — Scraper status, reports, trigger controls
Dashboard	`/dashboard`	Unchanged
Match Test	`/testing`	Unchanged

Test Coverage

SUITE	TESTS	STATUS
Unit — schemas (vitest)	45	✅ All passing
Unit — scoring (vitest)	41	✅ All passing
E2E — semantic search (vitest)	33	✅ All passing
Integration (vitest)	13	⚠️ Require running local server (see L016)
Standalone Total	**119**	✅

Note: Integration tests make live HTTP requests to `localhost:8090`. They pass when a local server is running but are excluded from CI. See LEARN.md L016 for details.

Security Posture

CONTROL	STATUS
Helmet security headers	✅ Active
Rate limiting (express-rate-limit)	✅ Active
CORS configuration	✅ Active (updated with scheduler headers)
Zod input validation	✅ All POST endpoints
npm audit	✅ 0 vulnerabilities

CONTROL	STATUS
No hardcoded secrets	✅ All via .env
Bearer token authentication	✅ **NEW** — All mutating endpoints protected
Optional auth on reads	✅ **NEW** — `req.authenticated` flag set
Scraper endpoint auth	✅ **NEW** — Cloud Scheduler header or API key

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Authentication System (NEW)

ASPECT	DETAIL
Strategy	Bearer token via `Authorization: Bearer` header
Protected endpoints	All mutating (POST, PATCH) except `/health`, `/match`, `/search/semantic`
Optional auth	All GET endpoints — works without token, sets `req.authenticated` flag
Token storage	`API_AUTH_TOKENS` env var (comma-separated, supports multiple tokens)
Scraper auth	`SCRAPER_API_KEY` env var or `X-CloudScheduler-JobName` header
Frontend	Login page stores token in `localStorage` as `cf-auth-token`
Dev mode	Auth bypassed when `NODE_ENV` is not `production`

Known Issues (Resolved vs. Remaining)

Resolved This Session

ISSUE (FROM V0.7.0)	RESOLUTION
React pages silently swallow API errors	✅ ErrorBoundary + Toast + ApiError class
CSS hardcoded values	✅ 17 new tokens, 21 RGBA replacements
No staging environment	✅ Deploy scripts + GitHub Actions CI/CD created
No user authentication	✅ Bearer token auth deployed to production

Remaining / New

ISSUE	SEVERITY	NOTES
Integration tests need running server	Medium	L016 — 13 tests require `localhost:8090`
Scraper trigger is a stub	Medium	Acknowledges request but doesn't run actual scraping
Scraper dashboard uses mock data	Low	Reports endpoint returns static fixture data
Cloud Monitoring not yet deployed	Medium	Scripts exist but alerts not yet pushed to GCP
Cloud Scheduler not yet deployed	Medium	Job configs exist but not submitted to GCP
Staging env not yet deployed	Medium	Scripts + CI/CD ready, needs `staging` branch push
Login is token-only	Low	No OAuth/password flow yet

ISSUE	SEVERITY	NOTES
Health endpoint still reports v0.7.0	Low	Version string in code needs bump

Recent Changes (2026-02-23)

COMMIT	CHANGE
`ccbc70b`	Add GitHub Actions staging deploy workflow
`d27cccf`	Update session handoff with LEARN.md updates and git activity
`e4c7529`	v0.8.0 sprint — error handling, auth, staging, monitoring, tests, LEARN.md global ref (43 files)

Deployment Info

FIELD	VALUE
Service	cf-matching-engine
Project	catchfire-app-2026
Region	us-central1
Revision	cf-matching-engine-00014-ln5
URL	https://cf-matching-engine-34240596768.us-central1.run.app
Last Deployed	2026-02-23
Deploy Command	`gcloud run deploy cf-matching-engine --source . --region us-central1 --project catchfire-app-2026`
Env Vars	`NODE_ENV`, `GCP_PROJECT`, `GCP_REGION`, `API_AUTH_TOKENS`, `SCRAPER_API_KEY`

Next Steps

PRIORITY	TASK	NOTES
High	Deploy Cloud Monitoring alerts	Run `scripts/deploy-monitoring.sh` — uptime + error rate + latency
High	Deploy Cloud Scheduler jobs	Run `scripts/deploy-scheduler.sh` — daily Vimeo/Behance, weekly all-platform
High	Deploy staging environment	Push `staging` branch to trigger GitHub Actions workflow
Medium	Wire scraper trigger to actual scraping logic	Currently a stub — needs cf-creator-database integration
Medium	Connect scraper dashboard to real Firestore data	Replace mock reports with `scraper_runs` collection
Medium	Bump version to v0.8.0 in health endpoint	Update version string in `src/index.js`
Low	Add OAuth/password auth flow	Current login is token-only
Low	Add loading skeletons to React pages	Replace "Loading..." text with skeleton components

PRIORITY	TASK	NOTES
Low	Fix 13 integration tests for CI	Need mock server or auto-start in test setup

Future Enhancements

PHASE	ENHANCEMENT	DESCRIPTION
8.8	Interactive Kanban Board	Real-time task tracking at `/app/kanban` with drag-and-drop, agent assignment, Firestore-backed persistence
9.1	Automated Scraper Pipeline	Full end-to-end: scheduler fires → scraper runs → data imported → embeddings generated → index updated
9.2	Multi-User Auth	Role-based access (admin, viewer), Google OAuth SSO
9.3	Creator Analytics	Trend tracking, engagement metrics over time, portfolio freshness scoring
9.4	Brief History & Saved Searches	Users can save searches, view past briefs, share results
9.5	Webhook Notifications	Slack/email alerts for new creator imports, scraper completions, match quality changes

Session Statistics

METRIC	VALUE
Tasks completed	14 / 14
Files modified	17 (this repo) + 5 (other repos)
Files created	26 (this repo) + 2 (global)
Tests added	41 (33 E2E + 8 fixture integrity)
CSS tokens added	17
RGBA replacements	21
New API endpoints	3
New React pages	2
Deploy scripts created	6
LEARN.md entries added	6 project (L011–L016) + 9 global (G001–G009)
Repos committed & pushed	6
Production revision	00011 → 00014

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